CampusCart AI-Enabled College Peer-to-Peer Marketplace

Complete Technology Stack & Enterprise Architecture Document

Target Institution: G.L. Bajaj Institute of Technology & Management (GLBITM)

Author: Engineering Core Team - Date: October 2026 - Version: 1.0.0 Production

1. SYSTEM ARCHITECTURE & REPOSITORY TOPOLOGY

CampusCart is architected as a high-performance monorepo with an AJIO-inspired e-commerce frontend, a resilient TypeScript Express micro-backend, dual Supabase PostgreSQL cloud storage, and Firebase OAuth.

FRONTEND SPA

- React 18 & TypeScript 5
- Vite 5 Build Engine
- Tailwind CSS 3.4 Design
- Lucide Iconography Suite

BACKEND API

- Node.js 20+ & Express.js
- TypeScript Strict Mode
- Multer Multipart Media
- Vercel Serverless Function

CLOUD & SECURITY

- Supabase Cloud PostgreSQL
- Supabase Cloud Storage
- Firebase Google Workspace
- Gemini 2.0 Multi-Modal AI

2. FRONTEND TECHNOLOGY SPECIFICATIONS

Layer / Component	Technology Choice	Version	Key Architectural Purpose
UI Framework	React	*v18.3.1*	Component hierarchy, virtual DOM, fast declarative rendering
Language	TypeScript	*v5.7.3*	Complete end-to-end static type safety, DTO interface contracts
Build Tooling	Vite	*v5.4.14*	Sub-second Hot Module Replacement (HMR) & Rollup tree-shaking
Styling System	Tailwind CSS	*v3.4.17*	Utility-first design tokens with AJIO-grade custom dark mode
CSS Post-Processor	PostCSS & Autoprefixer	*v8.5.1*	Cross-browser vendor prefixing and CSS modern standard parsing
Iconography Suite	Lucide React	*v0.475.0*	Tree-shakable clean SVG icons (ShieldCheck, MapPin, Cart, etc.)
Client Storage SDK	@supabase/supabase-js	*v2.118.0*	Direct client-side synchronization and asset URL generation
Auth Client SDK	Firebase Auth Web	*v13.1.0*	OAuth2 popup flow, Google Workspace SSO & token validation
State Management	React Context API	*Native 18*	Zero-dependency global stores: AuthContext, CartContext, Wishlist

3. BACKEND & API RUNTIME ARCHITECTURE

Layer / Component	Technology Choice	Version	Key Architectural Purpose
Server Runtime	Node.js (LTS)	*v20.x / 22.x*	Asynchronous non-blocking event-driven request lifecycle
Web Framework	Express.js	*v4.21.2*	RESTful router endpoints, middleware pipelines, static uploads
TypeScript Runner	tsx & tsc	*v4.19.2*	Live fast development server watcher and production compilation
Schema Validation	Zod	*v3.24.1*	Runtime type inference, request body validation, schema sanitization
CORS Management	cors	*v2.8.5*	Cross-Origin Resource Sharing policy for localhost and Vercel domains
Multipart Uploads	Multer	*v1.4.5*	Disk & buffer multipart stream parser with MIME and 5MB size limits
Environment Manager	dotenv	*v16.4.7*	Secure configuration loader for cloud connection strings and secrets
Serverless Wrapper	Vercel Node Adapter	*api/index.ts*	Exports Express instance directly for zero-cold-start cloud execution

4. DATABASE & CLOUD STORAGE ARCHITECTURE

CampusCart features a dual-persistence high-resilience storage architecture:

1. Supabase PostgreSQL Cloud: Enterprise relational database hosting complete DDL schemas (001 and 002 migrations) including tables for colleges, departments, users, categories, products, condition audits, meetups, and messages.
2. Supabase Storage (S3-Compatible): Cloud object bucket ('product-images') with public CDN endpoints for student uploads.
3. Unified In-Memory Store Fallback: Local database store allowing instantaneous offline and demo operation with auto-sync.

5. AUTHENTICATION & ACCESS CONTROL (FIREBASE GOOGLE SSO)

Feature / Security Rule	Technical Implementation Mechanism
Google Workspace OAuth	Firebase Web Authentication SDK v13 with GoogleAuthProvider popup interface
College Domain Gating	Validates accounts matching 'admission_number@glbitm.ac.in' and '@gibitm.ac.in'
Fallback / Guest Auth	Permits verified '@gmail.com' student testing with auto-mapped admission roll numbers
Role-Based Access Control	Strictly separates 'STUDENT' permissions from 'COLLEGE_ADMIN' moderation permissions
Session Management	Local token caching with zero server session leakage and instantaneous client logout

6. AI PRODUCT CONDITION & RESTRICTED POLICY PIPELINE

Module Name	Technology & Core Engine	Functionality & Safety Verification
Image Quality Analyzer	Visual Heuristics + Gemini	Screens for camera motion blur, low light, glare, and cropped framing
Category Classifier	Computer Vision + Rules	Prevents category mismatches (e.g. household items in Engineering D
Consistency Checker	NLP Text-Image Matching	Cross-references seller textual condition claims with photographic evi
Policy Detector	Restricted Term Database	Flags exam materials, weapons, unapproved chemicals, or illicit subst
Condition Appraisal	Visual Surface Grading	Evaluates wear and scores item as NEW, LIKE_NEW, GOOD, or FAI
Decision Synthesizer	Algorithmic Score Model	Computes calibrated trust score (0-100) and routes to Admin queue

7. DEVOPS, HOSTING & CI/CD INFRASTRUCTURE

- Production Platform: Vercel Cloud (Unified Edge Network with automatic HTTPS and global caching).
- Monorepo Architecture: Root 'package.json' with npm workspaces ('client', 'server') and isolated build targets.
- Git Repository: Hosted at GitHub ('https://github.com/pranav080506/CampusCart.git') with automated branch deployment.
- Serverless Function: 'api/index.ts' automatically executes the Express API on Vercel without requiring a standalone VPS.
- Security Standard: Sensitive secrets (.env files, service keys) strictly isolated from version control via '.gitignore'.

Document Certified for Campus Deployment - GLBITM Academic Season 2026-2027